ELEC70076
EE9-FPN2-10

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE PART IV: MEng and ACGI

SUSTAINABLE ELECTRICAL SYSTEMS

Thursday, 9 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question 1 & 4 in Booklet A, Questions 2 & 3 in Booklet B.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri, G. Strbac

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Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) Explain why variable speed operation is desirable for a wind turbine generator.

[3]

b) Wind turbines extract energy by slowing down a moving column of air. Explain qualitatively the trade-off involved in this process to maximise energy extraction. There is no need to write any equation.

[2]

c) The hourly wind speed (in m/s) profile across the year at a particular location is modelled by Weibull probability distribution where the probability of experiencing wind speed less than v_w is given by:

$$\Phi(v_w) = 1 - e^{-\left(\frac{v_w}{7.5}\right)^{1.9}}$$

At this location, a 9.5 MW direct-drive wind turbine with a permanent magnet synchronous generator (PMSG) is installed. The wind turbine parameters are given in Table 1.1 and the density of air can be assumed to be 1.22 kg/m^3.

Length of turbine blades	r	95 m
Maximum power coefficient	C_{p-max}	0.48
Optimum tip speed ratio	λ_{opt}	8.2
Cut-in speed	v_{w-in}	3 m/s
Cut-out speed	v_{w-out}	23 m/s

Table 1.1: Parameters of the wind turbine

i) What is the probability of encountering wind speeds in the region where the wind turbine operates in maximum power extraction mode by adjusting the turbine rotational speed?

[2]

ii) Determine the range of rotational speed (in rpm) of the turbine required to extract maximum power from wind.

[2]

iii) How many poles are needed for the PMSG if the minimum allowable stator frequency is 7.5 Hz?

[1]

d) A PV array is formed with 5 parallel strings with each string having 6 PV modules in series with a string diode. Each PV module has a rated power of 330 W with 18%

efficiency under standard test conditions (STC). The location of this PV array has a latitude of 55°N.

i) What should be the tilt angle (in degrees with respect to the horizontal) of the PV modules at solar noon if the declination angle on a particular day is 16.5°?

[2]

ii) What is the maximum total power output of the PV array when the irradiance is 600 W/m^2 assuming the same efficiency as under STC?

[2]

iii) How much power can be generated by the PV array with an irradiance of 600 W/m^2 if two PV modules within the same string are completely shaded?

[1]

e) The current-voltage characteristic of an ideal solar PV cell under 1000 W/ m^2 irradiance and 40°C is given by

$$I_{PV} = 5.5 - I_0\left[e^{\frac{qV_{PV}}{kT}} - 1\right]$$

where $q = 1.6 \times 10^{-19}$ C, $k = 1.39 \times 10^{-23}$ J/K, $I_0 = 1 \times 10^{-12}$ A

i) Calculate the open circuit voltage of the PV cell at 40°C.

[2]

ii) Calculate the short circuit current.

[1]

iii) Calculate the approximate power drawn by a 0.3 Ω resistor connected across the PV cell.

[1]

iv) Would the power drawn by a 2 Ω resistor across the PV cell be larger or smaller than that in part (iii)?

[1]

2. Question on balancing, Voltage Control and Security of Supply

Part A

a) Explain ways of managing voltage within distribution networks.

[2]

b) What is the maximum length, L, of a single-phase low voltage overhead line that supplies a maximum demand of 255 Amps with 0.95 (inductive) power factor connected at the end 2? The overhead line end 1 is connected to a single-phase pole mounted distribution transformer where the supply voltage is 230 V. Overhead line resistivity and reactivity, given as a sum for phase and neutral conductors, are 0.68 and 0.58 Ohms/km, respectively. The maximum statutory voltage deviation in an 11 kV network is 6%. The answer should contain a sketch of the considered system with denoted relevant variables and corresponding positive direction where relevant.

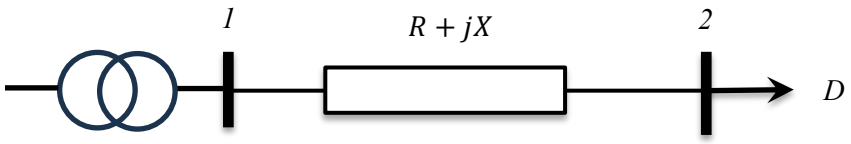


Figure 2.1: A transformer and a single-phase overhead line supplying a load.

[4]

Part B

A distribution network planner analyses the security of supply compliance of a substation, containing two 33/11 kV 7.5 MVA transformers, T1 and T2, at busbars 1, as shown in Figure 2.2, that supplies two demand centres, D1 and D2. Demand D1 and two non-intermittent generating units, G1 of 4 MW and G2 of 2 MW, are connected to busbars 1, while demand D2 and another non-intermittent generating unit, G3, of 2 MW are connected at busbars 2. Chronological daily profiles of demands D1 and D2 are shown in Figure 2.3, and the availability of non-intermittent generators is shown in Figure 2.2.

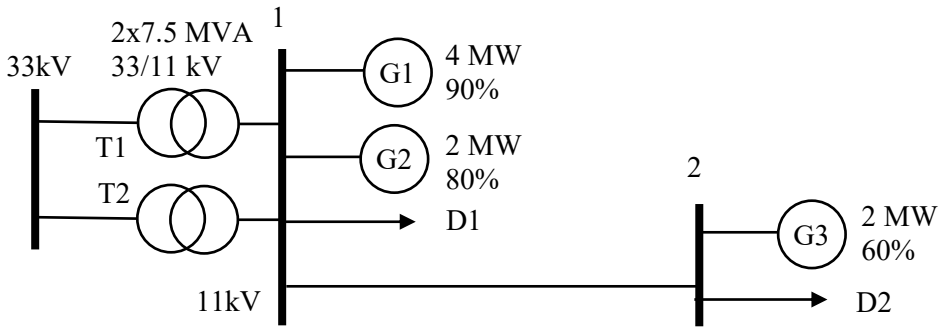


Figure 2.2 Diagram of the considered part of the network.

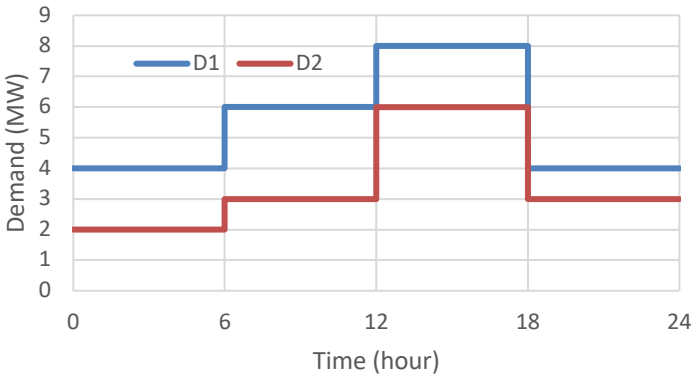


Figure 2.3 Daily demand profiles.

To help with solving the problem, an excerpt from the security of supply standard is provided in Table 2.1. For simplicity, assume unity power factor.

Table 2.1 Excerpt from the security of supply standard

Class of supply	Range of Group Demands	Minimum demand to be met after first circuit outage
A	Up to 1MW	In repair time: Group Demand
B	Over 1 to 12MW	(a) Within 3 hours: Group Demand Minus 1MW (b) In repair time: Group Demand
C	Over 12 to 60MW	(a) Within 15 minutes: Smaller of (Group Demand MINUS 12MW) and (2/3 Group Demand) (b) Within 3 hours Group Demand

a) What class of supply should be considered at busbar 1?

[1]

b) Form the capacity outage probability table for non-intermittent plants, G1, G2, and G3, providing security of supply support at busbars 1 by stating capacity in service and the corresponding probability value for each state.

[3]

c) Sketch the demand profile that will be used in the calculation of the contribution of generating units to the security of supply.

[1]

d) Calculate the Expected Energy Not Supplied by the system supplied by all three generation units using the demand profile sketched in c).

[4]

e) Calculate the capacity of the equivalent source, i.e. firm capacity, that never fails and results in the same Expected Energy Not Supplied.

[2]

f) Is the substation, containing two transformers: T1 and T2, at busbars 1 compliant with the security of supply standards? The cyclic rating factor of transformers is 1.3.

[3]

3. Question on Low Carbon Heat

Part A
Give a description of the following types of low carbon heating systems.

a) Air source heat pump

[2]

b) District heating

[2]

c) Hydrogen heating

[1]

d) Resistive heating

[1]

Part B
In 2008 the UK Government passed legislation entitled the Climate Change Act 2008. This required an 80% reduction in the UK's greenhouse gas emissions (GHG) by 2050 compared to emissions in 1990. The Government also established the Committee on Climate Change (CCC).

a) Briefly summarise the purpose of the CCC.

[1]

b) What amendment did the UK Government make in 2019?

[1]

Part C
A householder is trying to decide on whether to purchase a low-carbon resistive heating system or a more expensive air source heat pump system. The building has a floor area of 120 m^2 and an average thermal transmittance of 2.0 W_{th}/m^2K. In a typical year, the building's space heating is on for 150 days, 12 hours each day. The internal temperature is 18°C and the average external temperature when the space heating is on is 7°C.

a) Calculate the annual space heating demand in a typical year in MWh. State any assumptions made.

[4]

b) The cost of the heating systems are as follows:

	Heating system purchase and installation cost per annum (pa)	Running costs	Efficiency
Resistive heating	£750pa	0.3 £/kWh	100%
Air source heat pump	£1500pa	0.3 £/kWh	300%

i) What is the total annual cost of each system assuming space heat demand is as calculated in the above section (a) and which one should the householder select based on annual cost only?

[4]

ii) The householder is advised that there are further building insulation measures that can be taken to reduce space heat demand. How much does demand need to be reduced for the cost of each heating system to be the same?

[2]

iii) The householder does not understand how the ASHP system can have an efficiency in excess of 100%. Explain how this is possible.

[1]

iv) The householder has been advised to consider a hybrid ASHP as it may be more suited to his building. What is the difference between an ASHP and a hybrid ASHP?

[1]

4. Question on Demand Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

a) Explain the concept behind the Vehicle-to-Grid (V2G) technology and discuss the challenges and opportunities associated with using V2G in low-carbon electricity systems.

[2]

b) Baseline demand profile for the local distribution substation during evening peak hours is shown in Figure 4.1(a). Local Distribution Network Operator (DNO) has recently had to accommodate additional demand from 10 electric vehicles (EVs). The charging profile for a single EV is given in Figure 4.1(b). Answer the following questions:

i) What would be the new peak demand of the substation if all EVs start charging at 6 pm? Plot the new demand diagram with EV demand included.

[2]

ii) Assuming that each EV is flexible and can postpone its charging, what would be the earliest that all EVs can finish charging without increasing the total substation peak demand? What would be the new EV charging schedule, i.e., how many EVs should start charging at each hour? Plot the new diagram for total substation demand. (Assume that each EV can follow its own charging schedule.)

[6]

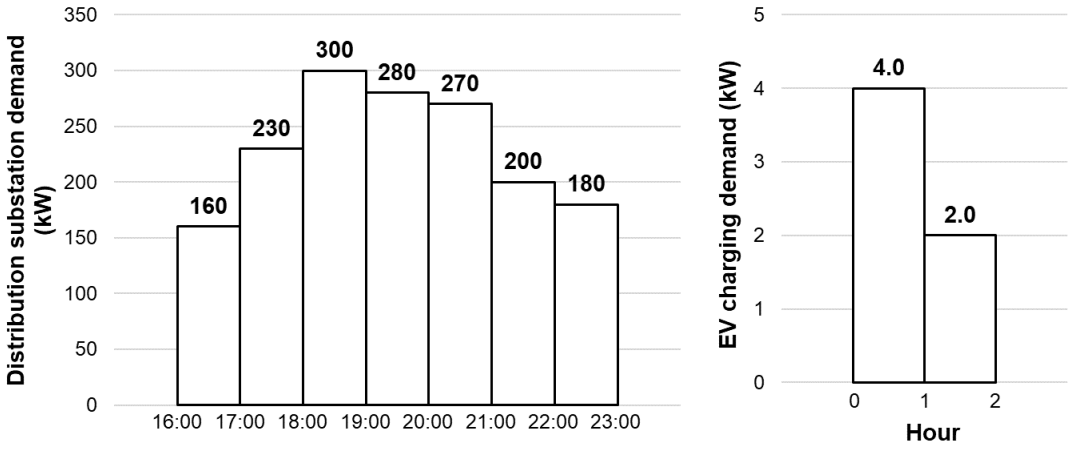


Figure 4.1 (a) Substation demand profile; (b) single EV charging demand profile

Hydro Power

c) A hydropower station with a reservoir has an effective head of 90 m. The height of the surface of the headwater in the reservoir is 140 m. There are 25 generators each rated at 600 MW, 20 kV, 50 Hz with a rated speed of 80 rpm.

i) What is the height (in m) of the surface of the tailwater?

[1]

ii) How many poles are there in each generator?

[1]

iii) What type of generator rotor construction is necessary to support the poles and how is it different from the rotor of thermal generators?

[2]

iv) How much discharge (in m^3/s) of water is necessary to produce the rated power of the hydropower station assuming 90% efficiency of the generator?

[1]

Offshore Transmission

d) Answer the following questions in the context of transmission of bulk power transmission from remote offshore wind farms through high-voltage subsea cables.

i) Why is HVDC preferred over AC?

[2]

ii) Why is VSC-HVDC preferred over LCC-HVDC?

[3]